

VPC ENDPOINTS

By Pelin Ece Burgun | September 2026

vpce-0fb28ebd663cf683c / NextWork VPC Endpoint			
Ayrıntılar	Özellikler	Uç Nokta	Ekler
Ayrıntılar Uç nokta ID'si vpce-0fb28ebd663cf683c VPC ID'si vpc-0a1ceabe313476c7f (NextWork-vpc) DNS kaydı IP türü service-defined Private DNS specified endpoint	Durum Mevcut Durum mesajı - IP adresi türü ipv4	Oluşturma zamanı 19 Eylül 2026 Cumartesi 13:11:33 GMT+3 Hizmet adı com.amazonaws.eu-central-1.s3 Service region eu-central-1	Uç nokta türü Gateway Özel DNS adları etkin Hayır Private DNS preference -

Introducing Today's Project!

What is Amazon VPC?

Amazon Virtual Private Cloud (VPC) lets you provision a logically isolated section of the AWS Cloud where you can launch resources in a virtual network that you define. It is useful for maintaining complete control over your network environment, including IP address ranges, subnets, route tables, and security settings.

How I used Amazon VPC in this project

In today's project, I used Amazon VPC to create a secure, isolated network foundation for my EC2 instance. The core focus was setting up a VPC Gateway Endpoint to establish a direct, private connection between my VPC and an S3 bucket. This ensured that my EC2 instance could communicate with S3 securely without the data traffic ever leaving the AWS network or going over the public internet.

One thing I didn't expect in this project was...

I didn't expect how strictly the S3 bucket policy would lock down access! Once I applied the policy restricting access strictly to the VPC endpoint, I was completely locked out of managing or deleting the bucket through the regular AWS Management Console. I had to rely entirely on the AWS CLI from my EC2 instance to empty and delete the bucket, which was a great hands-on lesson in security.

This project took me...

It took me about 1.5 hours to complete. Taking the time to properly troubleshoot the route table and test the endpoint policies with the AWS CLI took a bit of extra time, but it really helped solidify the concepts.

In the first part of my project...

Step 1 - Architecture set up

Setting up the foundational architecture for the project, which includes creating a VPC, launching an EC2 instance, and setting up an S3 bucket.

Step 2 - Connect to EC2 instance

Connecting directly to the EC2 instance using EC2 Instance Connect and attempting to access the S3 bucket through the public internet to see how it fails without credentials.

Step 3 - Set up access keys

Creating IAM Access Keys so the EC2 instance has the necessary credentials and permissions to interact with AWS services like S3.

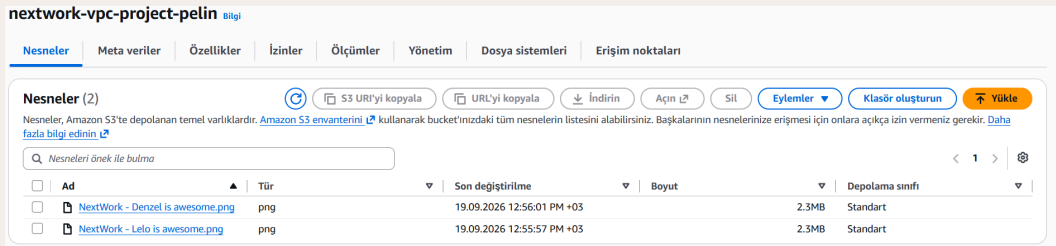
Step 4 - Interact with S3 bucket

Configuring the AWS CLI inside the EC2 instance using the newly created access keys and successfully interacting with the S3 bucket.

Architecture set up

A VPC, a public subnet, a t2.micro EC2 instance, a security group (allowing SSH), and a general-purpose S3 bucket.

I created a globally unique S3 bucket and uploaded two test files from my local computer into it.



Access keys

Credentials

AWS Access Key ID

AWS Secret Access Key

Default region name

Default output format

Access keys are long-term credentials for an IAM user or the AWS CLI. They consist of an access key ID (like a username) and a secret access key (like a password) used to authenticate programmatic requests to AWS services.

It is a secure string that pairs with your access key ID to authenticate your requests. It is only shown once upon creation and must be kept secret to prevent unauthorized access to your account.

Best practice

The best practice is to create an IAM role with the necessary permissions and attach that role directly to the EC2 instance, allowing it to inherit permissions securely without hardcoding keys.

Connecting to my S3 bucket

aws s3 ls (to list buckets) and then aws configure (to set up credentials).

It responded with a list of all the S3 buckets in my AWS account, including the nextwork-vpc-project bucket I just created.

```
[ec2-user@ip-10-0-9-51 ~]$ aws s3 ls
2026-09-19 09:54:25 nextwork-vpc-project-pelin
[ec2-user@ip-10-0-9-51 ~]$
```

Connecting to my S3 bucket

aws s3 ls s3://nextwork-vpc-project-[yourname]

```
[ec2-user@ip-10-0-9-51 ~]$ aws s3 ls s3://nextwork-vpc-project-pelin
2026-09-19 09:56:01    2431554 NextWork - Denzel is awesome.png
2026-09-19 09:55:57    2399812 NextWork - Lelo is awesome.png
[ec2-user@ip-10-0-9-51 ~]$
```

Uploading objects to S3

```
sudo touch /tmp/nextwork.txt
```

```
aws s3 cp /tmp/nextwork.txt s3://nextwork-vpc-project-[yourname]
```

(This command copied the newly created empty text file from the EC2 instance into the S3 bucket.)

```
aws s3 ls s3://nextwork-vpc-project-[yourname]
```

(This command listed the contents of the S3 bucket again to confirm that the nextwork.txt file was successfully uploaded.)

```
ec2-user@ip-10-0-9-51 ~]$ aws s3 ls s3://nextwork-vpc-project-pelin
2026-09-19 09:56:01      2431554 NextWork - Denzel is awesome.png
2026-09-19 09:55:57      2399812 NextWork - Lelo is awesome.png
ec2-user@ip-10-0-9-51 ~]$ sudo touch /tmp/nextwork.txt
ec2-user@ip-10-0-9-51 ~]$ aws s3 cp /tmp/nextwork.txt s3://nextwork-vpc-project-pelin
upload: ../../tmp/nextwork.txt to s3://nextwork-vpc-project-pelin/nextwork.txt
ec2-user@ip-10-0-9-51 ~]$ aws s3 ls s3://nextwork-vpc-project-pelin
2026-09-19 09:56:01      2431554 NextWork - Denzel is awesome.png
2026-09-19 09:55:57      2399812 NextWork - Lelo is awesome.png
2026-09-19 10:06:25           0 nextwork.txt
ec2-user@ip-10-0-9-51 ~]$
```

In the second part of my project...

Step 5 - Set up a Gateway

Creating a VPC Gateway Endpoint specifically for S3 to establish a secure, private connection between the VPC and the S3 service.

Step 6 - Bucket policies

Applying a strict bucket policy to the S3 bucket that denies all access unless the traffic comes explicitly from the VPC endpoint we just created.

Step 7 - Update route tables

Testing our VPC endpoint setup and troubleshooting an "Access Denied" error by checking and updating the VPC route table.

Step 8 - Validate endpoint connection

Testing the EC2 to S3 connection one last time to ensure it works, and experimenting with Endpoint Policies to see how they can be used to restrict access.

Setting up a Gateway

A Gateway is a specific type of VPC endpoint used exclusively for Amazon S3 and DynamoDB. It works by adding a route directly to your VPC route table that directs traffic bound for these services to the Gateway, keeping the connection private instead of routing it through the public internet.

What are endpoints?

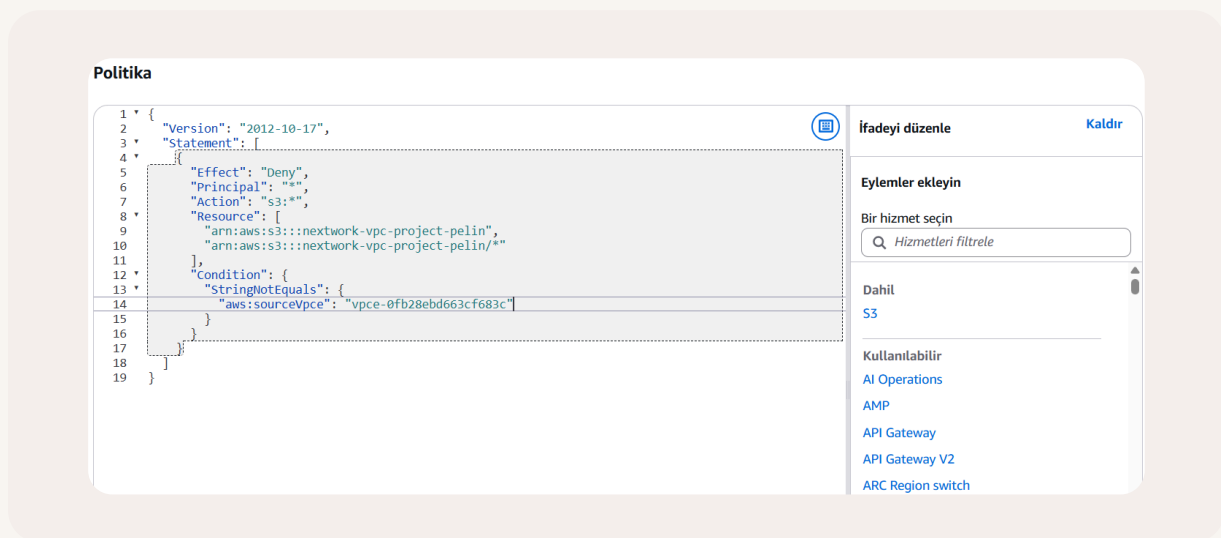
A VPC endpoint enables private connections between your VPC and supported AWS services without requiring that your traffic goes over the public internet, NAT devices, or VPN connections.

vpce-0fb28ebd663cf683c / NextWork VPC Endpoint			
Ayrıntılar	TOPLAM KULLANIM DURUMU	FATURA	ETKİLEMLER
Ayrıntılar			
Uç nokta ID'si vpce-0fb28ebd663cf683c	Durum Mevcut	Oluşturma zamanı 19 Eylül 2026 Cumartesi 13:11:33 GMT+3	Uç nokta türü Gateway
VPC ID'si vpc-0a1ceabe313476c7f (NextWork-vpc)	Durum mesajı -	Hizmet adı com.amazonaws.eu-central-1.s3	Özel DNS adları etkin Hayır
DNS kaydı IP türü service-defined	IP adresi türü ipv4	Service region eu-central-1	Private DNS preference -

Bucket policies

An IAM resource-based policy attached directly to an S3 bucket that defines access permissions—specifying exactly who can access the bucket and what actions they can perform.

The bucket policy denies all actions (s3:*) on the S3 bucket to everyone (Principal: "*"), unless the traffic originates from the specific VPC endpoint ID (aws:sourceVpce). This locks down the bucket so tightly that even AWS Management Console access is blocked.



The screenshot displays the AWS IAM console's 'Politika' (Policy) editor. The main area shows a JSON policy document with the following structure:

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Effect": "Deny",
6       "Principal": "*",
7       "Action": "s3:*",
8       "Resource": [
9         "arn:aws:s3:::network-vpc-project-pelin",
10        "arn:aws:s3:::network-vpc-project-pelin/*"
11      ],
12       "Condition": {
13         "StringNotEquals": {
14           "aws:sourceVpce": "vpce-0fb28ebd663cf683c"
15         }
16       }
17     }
18   ]
19 }
```

On the right side, there are two panels. The top panel, 'İfadeyi düzenle' (Edit expression), has a 'Kaldır' (Remove) button. The bottom panel, 'Eylemler ekleyin' (Add actions), includes a search bar 'Bir hizmet seçin' (Select a service) with the placeholder 'Hizmetleri filtrele' (Filter services). Below the search bar, a list of services is shown under the heading 'Dahil' (Included):

- S3
- Kullanılabilir
- AI Operations
- AMP
- API Gateway
- API Gateway V2
- ARC Region switch

Bucket policies

Because the bucket policy denies all traffic—even traffic coming from my own AWS Management Console—if it doesn't originate from the specific VPC endpoint.

Because the EC2 instance was still trying to reach S3 via the public internet. I needed to add a route that explicitly directs S3-bound traffic to the new VPC Gateway Endpoint.

Herkese açık erişimi engelle (bucket ayarları)

Düzenleyin

Herkese açık erişim, erişim denetim listeleri (ACL'ler), bucket politikaları, erişim noktası politikaları veya bunların tümü aracılığıyla bucket ve nesnelere verilir. Tüm S3 bucket ve nesnelerine herkese açık erişimin engellendiğinden emin olmak için, Tüm herkese açık erişimi engelle seçeneğini etkinleştirin. Bu ayarlar yalnızca bu bucket ve erişim noktalarına uygulanır. AWS, tüm herkese açık erişimi engelle seçeneğini etkinleştirmenizi önerir ancak bu ayarlardan herhangi birini uygulamadan önce uygulamalarınızın herkese açık erişim olmadan düzgün çalıştığından emin olun. Bu bucket'a veya içindeki nesnelere bir miktar herkese erişim sağlamak istiyorsanız, aşağıdaki ayarları tek tek özel depolama kullanım durumunuza uyacak şekilde özelleştirebilirsiniz. [Daha fazla bilgi edinin](#)

❌ Erişim reddedildi

▼ API yanıtı

User: arn:aws:iam::825765392980:user/Pelin-IAM-Admin is not authorized to perform: s3:GetBucketPublicAccessBlock on resource: "arn:aws:s3:::network-vpc-project-pelin" with an explicit deny in a resource-based policy

Bucket politikası

DüzenleyinSil

JSON ile yazılmış bucket politikası, bucket'ta saklanan nesnelere erişim imkanı sağlar. Bucket politikaları, diğer hesaplara ait nesneler için geçerli değildir. [Daha fazla bilgi edinin](#)

❌ Bucket politikasını alma izniniz bulunmuyor

s3:GetBucketPolicy izni almak için sizin tarafınızdan veya AWS yöneticiniz tarafından IAM izinleri güncellenmelidir. Gerekli izni aldıktan sonra sayfayı yenileyin. [Amazon S3'te kimlik ve erişim yönetimi](#) hakkında daha fazla bilgi edinin

▶ API yanıtı

Route table updates

I went to the VPC Endpoints console, selected my S3 Gateway endpoint, and clicked "Manage route tables". Then, I selected the checkbox for my public subnet's route table and clicked "Modify route tables" to automatically add a route that directs S3-bound traffic to the Gateway endpoint.

It successfully listed the objects inside the S3 bucket, proving the private endpoint connection works.

subnet-0911683d072e74c73 / NextWork-subnet-public1-eu-central-1a

Yönlendirme tablosu: rtb-07f5e558b066c0c73 / NextWork-rtb-public

Yönlendirme tablosu ilişkilendirmesi düzenle

Yönlendirmeler (3)

Yönlendirmeleri filtrele

Hedef	Hedef
10.0.0.0/16	local
0.0.0.0/0	igw-0d2aa961c7bc7da79
pl-6ea54007	vpce-0fb28ebd663cf683c

Endpoint policies

An IAM resource policy attached to a VPC endpoint that controls which AWS principals can use the endpoint to access the service, and what specific actions they are allowed to perform.

I edited the endpoint policy and changed the "Effect": "Allow" statement to "Effect": "Deny". The immediate effect was that it completely blocked the EC2 instance from accessing the S3 bucket, demonstrating how endpoint policies can quickly restrict access.

